

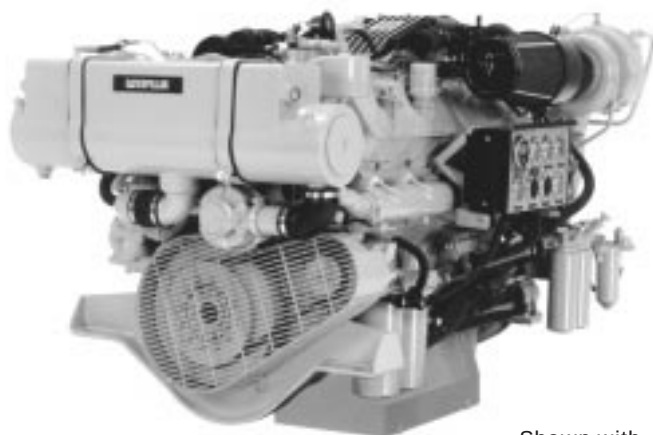
# CATERPILLAR®

## Marine Engine

# 3412C

1007 kW/1350 bhp

2300 rpm



Shown with  
Accessory Equipment

### STANDARD EQUIPMENT

#### Air Intake

aftercooler core, air cleaner (closed crankcase fumes disposal system)

#### Cooling

oil cooler, thermostats, sea water and jacket water pumps

#### Exhaust

watercooled manifold and turbocharger

Flywheel and Housing SAE No. 1

#### Fuel

priming and transfer pumps, filter, flexible fuel lines

#### Governor

hydramechanical

#### Instruments and gauges

fuel pressure, oil pressure, water temperature, service meter

*Power produced at the flywheel will be within standard tolerances up to 50° C (122° F) combustion air temperature measured at the air cleaner inlet, and fuel temperature up to 52° C (125° F) measured at the fuel filter base. Power rated in accordance with NMMA procedure as crankshaft power. Reduce crankshaft power by 3% for propeller shaft power.*

### CATERPILLAR® ENGINE SPECIFICATIONS

V-12, 4-Stroke-Cycle-Diesel

Bore—mm (in) ..... 137 (5.4)

Stroke—mm (in) ..... 152 (6.0)

Displacement—L (cu in)..... 27 (1649)

Rotation (from flywheel end) .. Counterclockwise

Compression Ratio ..... 14.1:1

Capacity for Liquids—L (U.S. gal)

Cooling System (engine only) ..... 83 (22)

Lube Oil System (refill) ..... 72 (19.0)

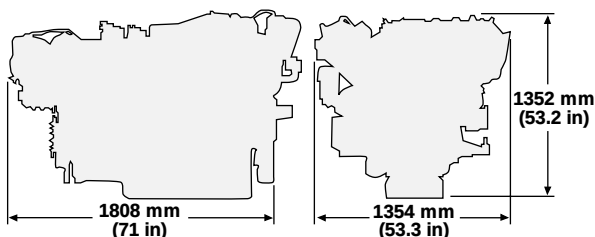
Oil Change Interval..... 19 000 (5000)/fuel

Engine Weight, Net Dry (approx)—

kg (lb) ..... 2313 (5100)

Governor..... Electronic

### DIMENSIONS



### RATING DEFINITIONS AND CONDITIONS

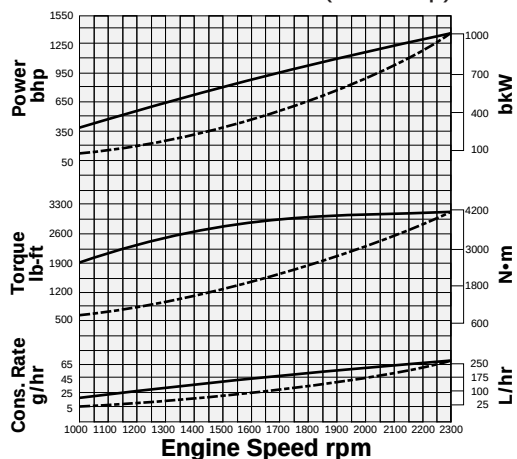
**Ratings** are based on SAE J1228/ISO8665 standard conditions of 100 kPa (29.61 in Hg) and 25° C (77° F). These ratings also apply at ISO3046/1, DIN6271, and BS5514 conditions of 100 kPa (29.61 in Hg), 27° C (81° F), and 60% relative humidity.

**Fuel rates** are based on fuel oil of 35° API [16° C (60° F)] gravity having an LHV of 42 780 kJ/kg (18 390 Btu/lb) when used at 29° C (85° F) and weighing 838.9 g/liter (7.001 lbs/U.S. gal).

Additional ratings may be available for specific customer requirements. Consult your Caterpillar representative for additional information.

## PERFORMANCE CURVES

**E<sub>hp</sub> Rating – 2300 rpm**  
1007 bkW (1350 bhp) 1369 mhp



- Prop Demand - - - - - 3.0 Exponent
- Engine Performance Parameters: Power +/- 3%;  
Specific Fuel Consumption +/- 3%; Fuel Rate +/- 5%.

Prop Demand Curve Data

| Speed rpm | Power bkW | Torque N•m | Fuel Cons g/kW-hr | Fuel Rate L/hr |
|-----------|-----------|------------|-------------------|----------------|
| 2300      | 1007      | 4181       | 221               | 265.3          |
| 2200      | 881       | 3825       | 213               | 223.4          |
| 2000      | 662       | 3161       | 208               | 164.3          |
| 1800      | 483       | 2560       | 214               | 122.9          |
| 1600      | 339       | 2023       | 222               | 89.5           |
| 1400      | 227       | 1549       | 231               | 62.5           |
| 1200      | 143       | 1138       | 243               | 41.4           |
| 1000      | 83        | 790        | 254               | 25.1           |

| Speed rpm | Power bhp | Torque lb-ft | Fuel Cons lb/bhp-hr | Fuel Rate g/hr |
|-----------|-----------|--------------|---------------------|----------------|
| 2300      | 1350      | 3084         | .363                | 70.1           |
| 2200      | 1182      | 2821         | .350                | 59.0           |
| 2000      | 888       | 2331         | .342                | 43.4           |
| 1800      | 647       | 1888         | .351                | 32.5           |
| 1600      | 455       | 1492         | .364                | 23.6           |
| 1400      | 305       | 1142         | .379                | 16.5           |
| 1200      | 192       | 839          | .399                | 10.9           |
| 1000      | 111       | 583          | .418                | 6.6            |

Max Power Curve Data

| Power bkW | Torque N•m | Fuel Cons g/kW-hr | Fuel Rate L/hr |
|-----------|------------|-------------------|----------------|
| 1007      | 4181       | 221               | 265.3          |
| 951       | 4129       | 214               | 242.1          |
| 854       | 4077       | 206               | 209.7          |
| 772       | 4097       | 205               | 188.9          |
| 679       | 4051       | 211               | 171.0          |
| 545       | 3720       | 222               | 144.0          |
| 392       | 3122       | 227               | 106.3          |
| 275       | 2629       | 213               | 70.0           |

| Power bhp | Torque lb-ft | Fuel Cons lb/bhp-hr | Fuel Rate g/hr |
|-----------|--------------|---------------------|----------------|
| 1350      | 3084         | .363                | 70.1           |
| 1276      | 3045         | .351                | 64.0           |
| 1145      | 3007         | .339                | 55.4           |
| 1036      | 3022         | .337                | 49.9           |
| 910       | 2988         | .347                | 45.2           |
| 731       | 2744         | .364                | 38.0           |
| 526       | 2303         | .374                | 28.1           |
| 369       | 1939         | .351                | 18.5           |

E<sub>hp</sub> RATING – for use in non-revenue producing vessels.